Solvency Field Theory

V10.8 Companion Note

Theorem 6b: Branch Consolidation, Count-Normalized Single-Field Reduction, and the Perturbative Gate

Status: Lead 6b branch consolidated. Background normalization and perturbative structural compatibility preserved. Full covariant and strong-field closure remain open.

Date: April 22, 2026

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1. Purpose of this note

V10.7 selected the lead minisuperspace branch of Theorem 6b: a **single-field, slaved/constrained, count-normalized branch** in which the source term drives the bookkeeping field and the count sector supplies the final $4\pi^2$ normalization. In that branch dust is exact and radiation is asymptotically exact in the background sector.

The next question was whether this same branch can also pass the **perturbative gate**. The target is not vague. V10.1 already fixed the perturbative source structure used in the CLASS implementation, and Theorem 6 requires any candidate action to reduce to that structure on FLRW.

This note records the result of that comparison. The conclusion is not that 6b is fully closed. The conclusion is narrower and more useful:

> **The lead single-field constrained/slaved branch is structurally compatible with the V10.1 perturbative source sector.**

The remaining perturbative work is now a normalization-and-translation task, not a new conceptual frontier.

2. What V10.7 already established

The lead 6b branch is the single-field constrained/slaved branch

$$S_{1\mathrm{f}} = \int d^4x\,\sqrt{-g}\,\Big[\frac{R}{16\pi G} - \frac{K}{2}\nabla_\mu\chi\nabla^\mu\chi - V_0 e^{-\chi} - \eta\,\nabla_\mu\chi\,J^\mu_{\mathrm{ops}} + \mathcal{L}_m + \Lambda_C\,\mathcal{C}[g,\chi,\psi]\Big],$$

with clock congruence

$$u^\mu = \frac{\nabla^\mu\chi}{\sqrt{-\nabla_\alpha\chi\nabla^\alpha\chi}}, \qquad h^{\mu\nu} = g^{\mu\nu} + \nu^\mu\nu^\nu,$$

and operation-density current

$$J^{\mu}_{\rm ops} = \rho_{\rm ops}\,\nu^{\mu}, \qquad \rho_{\rm ops} = \frac{T_{\mu\nu}\nu^{\mu}\nu^{\nu} + h^{\mu\nu}T_{\mu\nu}}{h}.$$

In the local clock-field frame,

$$\rho_{\rm ops} = \frac{\rho + 3p}{h}.$$

The sourced FLRW equation is

$$K(\ddot{\chi} + 3H\dot{\chi}) - V_0 e^{-\chi} = \eta \nabla_{\mu} J^{\mu}_{\rm ops}, \qquad \nabla_{\mu} J^{\mu}_{\rm ops} = -\frac{3Hw(1+3w)\rho}{h},$$

so dust gives zero driving and radiation gives active driving. The count-normalized slaved branch then produces

$$H^2 = \beta\,(\rho + V_{\rm eff})\,, \qquad \beta = \frac{1}{4\pi^2},$$

with dust exact and radiation asymptotically exact.

This is the background branch that the perturbative sector must now inherit.

3. Fixed perturbative target from V10.1

The perturbative gate is concrete because V10.1 already fixed the effective source structure used in the solver-level implementation.

At the minimal heating level,

$$\delta Q_{\gamma,k}(\eta) = \beta_Q(\eta)\,\frac{k^2}{a(\eta)^2}\,\delta\Phi_k(\eta).$$

At the CLASS insertion level,

$$S_{\rm SFT}(k,\tau) = A_{\rm sft}\,g(\tau)\,\beta_Q(\tau)\,\frac{k^2}{a(\tau)^2}\,\Phi_{\rm eq}(\tau)\,\eta_k\,W\!\left(\frac{k}{k_c(\tau)}\right).$$

So any 6b branch that claims perturbative compatibility must reproduce, at minimum:

1. the same phase-curvature operator k^2/a^2,
2. the same order-parameter logic $\Phi_{\rm eq}(\tau)$,
3. and a finite instability-band window $W(k/k_c)$.

The question is therefore not "does the action produce some perturbation?" The question is whether the lead 6b branch reduces to this already-fixed V10.1 structure.

4. Single-field perturbation ansatz

Linearize the lead branch around FLRW:

$$\chi(\eta, \mathbf{x}) = \bar{\chi}(\eta) + \delta\chi(\eta, \mathbf{x}).$$

In the single-field branch, the old two-field bridge is absorbed into the identification

$$\Phi = 2\pi\chi, \qquad \delta\Phi = 2\pi\,\delta\chi.$$

This is not an added dynamical assumption. It is the same branch-selection simplification already preserved in V10.7, now applied at perturbative level.

Retaining a small effective perturbative stiffness $K_{\rm pert}$ while keeping the background branch slaved, the linearized scalar equation has the schematic form

$$K_{\rm pert}\left(\delta\chi_k'' + 2\mathcal{H}\,\delta\chi_k' + c_\chi^2\frac{k^2}{a^2}\delta\chi_k\right) + M_{\rm eff}^2(\eta)\,\delta\chi_k = \eta\,\delta\Big(\nabla_\mu J^\mu_{\rm ops}\Big)_k + \text{metric-mixing terms}.$$

For the exponential maintenance potential,

$$V(\chi) = V_0 e^{-\chi}, \qquad M_{\rm eff}^2(\eta) = V_{,\chi\chi}(\bar{\chi}) = V_0 e^{-\bar{\chi}}.$$

The important fact is immediate: the perturbation operator contains the same gradient-curvature structure as V10.1.

5. Operator and variable match

The perturbative operator match is straightforward.

The lead branch produces

$$\frac{k^2}{a^2}\delta\chi_k.$$

Using

$$\delta\Phi_k = 2\pi\,\delta\chi_k,$$

this becomes

$$\frac{k^2}{a^2}\delta\chi_k = \frac{1}{2\pi}\,\frac{k^2}{a^2}\delta\Phi_k.$$

So at the operator level the single-field branch automatically reproduces the same phase-curvature source structure that V10.1 already used:

$$\delta Q_{\gamma,k} \propto \frac{k^2}{a^2}\delta\Phi_k.$$

This is the first major perturbative gate, and it is now cleanly passed.

6. Effective source-history dictionary

The perturbative question then reduces from "operator match" to "history match."

The V10.1 packaging uses

$$\beta_Q(\tau)\,\Phi_{\rm eq}(\tau)\,W\left(\frac{k}{k_c(\tau)}\right).$$

The lead 6b branch supplies the natural single-field dictionary

$$\Phi_{\rm eq}^{\rm eff}(\tau) = 2\pi\,\bar{\chi}(\tau),$$

with background slaved solution determined by

$$V_0 e^{-\bar{\chi}} = V_{\rm eff} = AH\rho, \qquad A = \frac{3\eta w(1+3w)}{h}.$$

For radiation with $\eta = 1$ and $h = 1$, this gives $A = 2$.

The V10.1-inspired Hubble-limited closure suggests the effective coefficient history

$$\beta_Q^{\rm eff}(\tau) \sim q_T f_b f_r\,\frac{\epsilon_\Phi}{H(\tau)}, \qquad \epsilon_\Phi = \frac{1}{4\pi^2}.$$

So the natural effective perturbative source in the lead 6b branch is

$$S_{\rm eff}(k,\tau) \sim \beta_Q^{\rm eff}(\tau)\,\frac{k^2}{a(\tau)^2}\,\Phi_{\rm eq}^{\rm eff}(\tau)\,W\left(\frac{k}{k_c^{\rm eff}(\tau)}\right).$$

This is not a re-invention of the V10.1 source. It is the same packaging read through the lead single-field branch.

7. Instability window and effective cutoff

A minimal structural estimate for the perturbative cutoff is

$$k_c^{\rm eff}(a) \sim \frac{a}{c_\chi}\sqrt{\frac{M_{\rm eff}^2(a)}{K_{\rm pert}}}.$$

This uses the same logic already implicit in the V10.1 instability-band construction: the perturbative source is active only where the lag/phase sector supports a real instability band.

In the single-field constrained branch, the diagnostic behavior is exactly what the perturbative gate requires:

- $k_c^{\rm eff}(a)$ is finite,
- it decreases strongly with time,
- lower-k and higher-k modes are separated by the window,

- each test mode develops a finite peak epoch,
- and the window shuts the source off at late times.

That is qualitatively the same role played in V10.1 by the instability-band window $W(k/k_c)$.

8. Why the source history is not pathological

At first glance, the background factors can look strange. In the radiation branch,

$$\beta_Q^{\text{eff}} \sim \frac{1}{H}, \qquad \Phi_{\text{eq}}^{\text{eff}} \sim 2\pi\bar{\chi}.$$

Since the count-constrained radiation branch gives asymptotically

$$H \propto a^{-2},$$

we get

$$\beta_Q^{\text{eff}} \propto a^2.$$

Meanwhile the order parameter grows only logarithmically through the slaved relation,

$$\Phi_{\text{eq}}^{\text{eff}} \sim \ln a.$$

So the background prefactor

$$\beta_Q^{\text{eff}} \Phi_{\text{eq}}^{\text{eff}}$$

does grow. But this is not the physical source by itself. The full source contains the explicit curvature operator

$$\frac{k^2}{a^2},$$

which cancels the a^2 growth of β_Q^{eff}, leaving only slow logarithmic evolution before the instability window suppresses the source altogether.

So the apparent growth in the prefactor history is not a pathology. It is the expected bookkeeping piece before multiplying by the curvature operator and the mode window.

9. What the perturbative diagnostic actually showed

The structural perturbative diagnostic established three things.

9.1. Background history is smooth

The effective history functions

$$\beta_Q^{\text{eff}}(a), \qquad \Phi_{\text{eq}}^{\text{eff}}(a)$$

are smooth and monotonic in the constrained branch. There is no singular or oscillatory behavior that would make the source packaging ill-defined.

9.2. Mode separation is real

The finite $k_c^{\rm eff}(a)$ produces distinct behavior for different representative modes. Lower-k and higher-k modes peak at different epochs, and all sampled modes are suppressed at late times by the instability window.

9.3. Finite-epoch source kernels appear naturally

The effective kernels

$$S_{\rm eff}(k,a) \sim \beta_Q^{\rm eff}(a)\,\frac{k^2}{a^2}\,\Phi_{\rm eq}^{\rm eff}(a)\,W\!\left(\frac{k}{k_c^{\rm eff}(a)}\right)$$

peak over finite intervals and then decay. This is exactly the qualitative behavior required by the V10.1 perturbative source chain.

10. What is already matched and what is not

The perturbative gate is now split cleanly into "already matched" and "still to verify."

10.1. Already matched in substance

1. **Variable match:**

$$\delta\Phi = 2\pi\,\delta\chi.$$

2. **Operator match:**

$$\frac{k^2}{a^2}\delta\chi_k \leftrightarrow \frac{k^2}{a^2}\delta\Phi_k.$$

3. **Instability-band logic:** finite $k_c^{\rm eff}(a)$ and mode windowing survive in the lead branch.

10.2. Still a verification task

The remaining perturbative job is not conceptual. It is the exact translation between the effective single-field history

$$\beta_Q^{\rm eff}(\tau)\,\Phi_{\rm eq}^{\rm eff}(\tau)$$

and the precise V10.1 implementation packaging

$$\beta_Q(\tau)\,\Phi_{\rm eq}(\tau)$$

including any normalization and visibility-factor bookkeeping.

That is why the right statement is now “structural compatibility” rather than “exact perturbative identity already proved.”

11. Updated status of the remaining open items

After V10.8 the open-item stack is sharper:

11.1. Perturbative gate

No longer a frontier problem. It is now a structural pass with remaining normalization translation.

11.2. Covariant single-field formalization

Still assembly work. The lead branch is identified, but the full local covariant constrained-action formulation remains to be written cleanly.

11.3. Count constraint

The minisuperspace count multiplier now looks like the FLRW shadow of the full count constraint. But the genuinely local covariant count relation remains to be derived.

11.4. Strong-field gate

Still the one gate most likely to surprise us. The structural argument for vacuum recovery is good, but the full static spherically symmetric equations for the constrained single-field action have not yet been solved.

12. Bottom line

The strongest V10.8 statement is:

> **The lead Theorem 6b branch is now the single-field, slaved/constrained, count-normalized branch, and that branch is structurally compatible with the V10.1 perturbative source sector. It reproduces the same phase-curvature operator, supports a finite instability-band window, and generates finite-epoch source kernels that decay at late times. The remaining perturbative work is exact normalization matching, not a new conceptual wall.**

This does not finish 6b. But it closes the branch-selection and perturbative-compatibility items strongly enough to promote them from working-note status to a V10.8-level companion note.